

Real-time Domain Adaptation in Semantic Segmentation

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Abstract

+ This project explores the problem of domain adaptation for real-time semantic segmentation in synthetic-to-real urban environments. The objective is to mitigate performance degradation when models are trained on synthetic datasets (e.g., GTA5) and evaluated on real-world datasets (e.g., Cityscapes). The BiSeNet architecture is adopted for its fast inference capabilities and efficiency. Initially, BiSeNet is trained on the Cityscapes dataset to establish an upper-bound performance benchmark. Subsequently, it is trained solely on GTA5, revealing a substantial performance drop due to domain shift. To address this, an image-level domain adaptation technique, Fourier Domain Adaptation (FDA), is applied to reduce the gap between source and target domains, yielding moderate improvements. Building on FDA, a self-training method known as Domain Adaptation via Cross-domain Mixing (DACS) is implemented using pseudo-labels generated from FDA-trained models. Furthermore, an extension to this pipeline introduces a hybrid segmentation loss function combining CrossEntropy and Dice Loss to better handle class imbalance, particularly for underrepresented classes such as bicycle and fence, without modifying the network architecture. The final end-to-end pipeline integrates dataset preprocessing, baseline training, domain adaptation, and novel loss formulation to enhance segmentation performance in real-time settings. The complete implementation of this project is available on GitHub at <https://github.com/safiullahnaseerm-stack/Semantic-Segmentation>.

1. Introduction

In the field of computer vision, Semantic segmentation is important in applications like autonomous driving, medical imaging and robotics [5]. With the recent developments of deep learning methods, the accuracy of the segmentation of annotated big sets of data has notably increased. Never-

theless, generalization of a model which was trained with respect to one domain tends to become poor when used on another domain as a result of domain shift. This presents significant difficulty in practice when few or costly data annotated with labels in the target domain are available.

In an attempt to address this problem, synthetic data like GTA5 is highly used in the context of their accessibility and affordability of annotation. In spite of the clear visual quality of synthetic images, there has been the frequent observation that models trained on purely synthetic data do not perform very well when tested over real world datasets like Cityscapes. This decline of performance is explained by domain differences involving texture, light, and distribution of objects.

In order to fill this gap, the technique of domain adaptation (DA) in semantic segmentation has appeared. The aim of DA is to transfer knowledge residing in a source domain, which is labeled, to an unlabelled target domain by matching the distributions on the image, feature, or output level [6].

This project examines domain adaptation methodologies in real-time semantic segmentation of synthetic (GTA5) to real world (Cityscapes) images by using the BiSeNet framework. BiSeNet is chosen because it is lightweight and has a great inference speed, therefore, it is used to meet real-time processes. The research is organized into four key stages, dataset processing, baseline training, domain shift assessment, and domain adaptation. We investigate two main kinds of domain adaptation algorithms, Fourier Domain Adaptation (FDA) and Domain Adaptation via Cross-domain Mixing (DACS). The first of which executes a low-frequency translation of the images and the other one makes use of pseudo-labeling and classes mixing. Furthermore, in the last step, extension is proposed with an addition of hybrid segmentation loss that balances cross entropy loss and dice loss assigned to the class imbalance.

2. Related Work

Semantic segmentation's domain adaptation has become an important topic in research especially where there is a change of data on a move from synthetic to real data. This part gives an overview of the previous researches on semantic segmentation, domain shifts, domain adaptation methods, and networks of real-time segmentation.

2.1. Semantic Segmentation and Domain Shift

Semantic segmentation as a task of computer vision is the process of assigning a label describing the class to each pixel of image [1]. In the recent past, deep learning has resulted in tremendous breakthroughs with regard to segmentation, especially through convolutional neural networks (CNNs). DeepLabV2 and BiSeNet are the networks that have facilitated advancement [12]. Performance may however erode when these models are used across domains (the domain shift problem) because of visual, textural and distributional differences. Following image shows semantic segmentation.



Figure 1. Semantic Segmentation (online available at: <https://pub.towardsai.net/machine-learning-23997460cbc4>).

Hence, there is an increased usage of synthetic datasets such as GTA5, which are available as training data, as pixel-wise annotated real-world datasets such as Cityscapes are too expensive to collect. However, the problem of domain shift between these two datasets makes model generalisation poor, and thus should be addressed through domain adaptation approaches to fill that gap.

2.2. Domain Adaptation Strategies

Domain adaptation (DA) is a technique that bothers with the domain shift problem by matching the distributions of the source domain and the target domain. DA approaches may be broadly divided into three groups, including adversarial, image-to-image translation, and self-training.

Image-to-Image Translation: This approach is based on translating the images of the source domain to make them similar as the images of the target domain [7]. An example technique is Fourier Domain Adaptation (FDA) in which low-frequency parts of source images are replaced with those of the target images in an effort to transform visual appearance without varying content [13].

Self-Training and Pseudo-Labeling: The techniques involved are training with predictions on the target domain as pseudo-labels. DACS processes pseudo-labelling with robust learning because DACS mixes strong augmentation of source image with weak augmentation of target image [11].

2.3. Real-time Semantic Segmentation Networks

Real-time Semantic segmentation models must balance accuracy and computational efficiency, especially in real-time applications.

DeepLabV2 is a widely used architecture known for its high segmentation accuracy. It employs atrous spatial pyramid pooling (ASPP) and fully connected Conditional Random Fields (CRFs) in order to refine boundaries and capture contextual information [2]. However, its computational cost and inference time make it less suitable for real-time tasks.

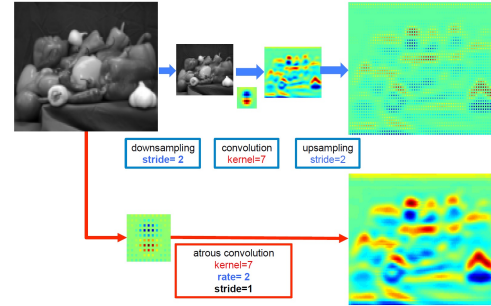


Figure 2. DeepLabV2 architecture illustrating the use of atrous convolution and Atrous Spatial Pyramid Pooling (ASPP)

BiSeNet is one of the real-time segmentation networks, which aims at fast inference. It uses a high-resolution feature path and a semantic richness path, both of which include a spatial path and a context path identifying it as an appropriate architecture when resources are limited [15].

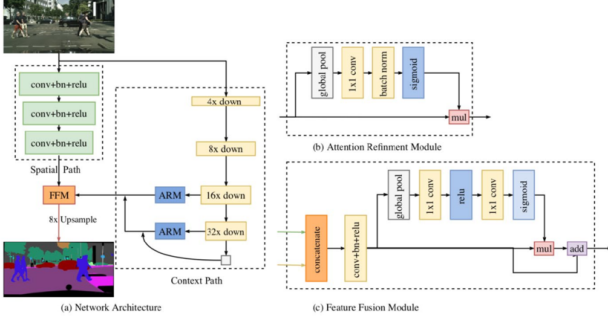


Figure 3. Detailed architecture of BiSeNet.

2.4. Hybrid Loss Functions for Imbalanced Classes

In addition to that, Class imbalance is still a problem to deal with in semantic segmentation, and it is harder to solve with the poorly represented classes like “bicycle” or “fence” [8]. Compositions of CrossEntropy Loss and Dice Loss known as hybrid loss functions have been experimented to enhance the accuracy of the segmentation on small classes. Dice Loss is complementary to CrossEntropy loss through being invariant to the overlap area between regions an advantage in situations where there is extreme skew in the number of classes present [14].

Hence, the current project supports the existing literature on semantic segmentation and domain adaptation by extending BiSeNet with FDA and DACS. Moreover, it also offers a hybrid loss to cope with the imbalance of classes.

2.5. Evaluation Metrics: mIoU, FLOPs, and Parameters

To assess the performance of segmentation models and domain adaptation strategies, multiple evaluation metrics are considered.

Mean Intersection over Union (mIoU) is the most commonly used metric for semantic segmentation [9]. It measures the overlap between the predicted and ground truth segments across all classes. Mathematically, for a class i , the IoU is calculated as:

$$\text{IoU}_i = \frac{TP_i}{TP_i + FP_i + FN_i}$$

where TP_i , FP_i , and FN_i are the true positives, false positives, and false negatives for class i , respectively. The mean is taken over all C classes:

$$\text{mIoU} = \frac{1}{C} \sum_{i=1}^C \text{IoU}_i$$

FLOPs (Floating Point Operations) are used to estimate the computational cost of a model. It represents the total number of operations required for a forward pass. Mod-

els with fewer FLOPs are generally faster and more efficient [4].

Number of Parameters indicates the model size and memory footprint. Smaller models with fewer parameters are easier to deploy in real-time systems and on devices with limited resources [3].

In this project, mIoU is used as the primary evaluation metric for segmentation accuracy, while FLOPs and parameter count are referenced to compare model complexity between BiSeNet and DeepLabV2.

3. Methodology

3.1. Step 1: Baseline on Target Domain

A baseline of models that are trained and tested only on the Cityscapes data are used to provide an upper-bound reference on performance, since this is the target real-world domain. Doing this allows evaluating how accurate we can get without domain adaptation, which will be followed by measuring the performance of domain adaptation techniques.

Two semantic segmentation architectures were explored for this purpose:

- **DeepLabV2:** A conventional yet strong encoder-decoder scheme that is characterized by the use of atrous spatial pyramid pooling (ASPP) to deal with multi-scale context.
- **BiSeNet (Bilateral Segmentation Network):** The dual-path architecture which efficiently creates balance between spatial detailing and contextual semantics. BiSeNet is chosen as it is efficient in the information inference and can be successfully applied to real-time urban scene perception.

Training and evaluation on both models was done on the Cityscapes training set and Cityscapes validation set respectively. When training, generalisation was increased using standard forms of data augmentations, i.e. random cropping and flipping. Its performance was subsequently gauged by standard segmentation algorithms such as mean Intersection-over-Union (mIoU), the number of parameters and the number of floating-point operations (FLOPs).

3.2. Step 2: Domain Shift Analysis

In order to study the impacts of domain shift, BiSeNet is trained on synthetic GTA5 only and evaluated on the Cityscapes validation set without adaptation. This environment models a scenario of a real world application, with annotated data in a synthetic space because the goal is to check that whether the model is able to generalise the live city or real urban conditions.

The outcome demonstrated a significant decrease in performance with a Cityscapes baseline. This decrease in performance is due to the domain shift. Domain shift refers to the discrepancy in the data distribution in both the source domain (GTA5) and target domain (Cityscapes). Causes of this change are variations in visual style, resolution, weather conditions, scene arrangement and sensor typography. Even though the images offered by GTA5 are highly annotated, the artificial rendering does not offer all the complexity and noise of real-life imagery.

This analysis stresses the necessity of domain adaptation methods to reduce the gap and enhance the model generalization during real deployment conditions.

3.3. Step 3: Data Augmentation

To reduce the possible overfitting on the synthetically generated images and enhance learning a better model generalization a number of data augmentating procedures are followed on the GTA5 data set before undergoing training. Their choice of augmentation is to imitate the possible visual variations which can happen during the real driving conditions to enhance the model robustness to domain differences.

In this project, the following augmentations are implemented:

- **Horizontal Flipping:** This technique randomly flipped images which results in different driving perspectives and mirror variations [16].
- **Color Jitter:** This technique helped to introduce controlled changes in brightness, contrast, and saturation to mimic diverse lighting conditions [10].
- **Combination of Flip and Jitter:** Both augmentations techniques are combined to assess their joint impact on model performance.

All the augmentations are measured separately and together. The target augmentations techniques of quantitative methods are used to enhance mIoU results, especially on classes with high visual variations. This enhancement scheme developed stronger basis of feature learning before the highly designed domain adaptation strategies.

3.4. Step 4: Domain Adaptation

Though data augmentation had its benefits, domain shift still remained a significant obstacle. For this, two domain adaptation techniques are explored i.e. Fourier Domain Adaptation (FDA) and Domain Adaptation by Cross-domain Mixing (DACS). Both of them attempt to lessen the domain gap between synthetic data of GTA5 and real Cityscapes data, on both image and feature level.

Fourier Domain Adaptation (FDA): FDA is an image-to-image translation technique capable of working in the frequency domain. The concept is to learn to remove low-frequency information in real target (Cityscapes) images and place them in synthetic source (GTA5) images. This does not affect the structure of the source image but aligns the appearance statistics to that of the target domain.

Mathematically, the adapted source image I_{FDA} is defined as:

$$I_{FDA} = \mathcal{F}^{-1}((1 - M) \cdot \mathcal{F}(I_s) + M \cdot \mathcal{F}(I_t))$$

Where $\mathcal{F}$ and $\mathcal{F}^{-1}$ denote the Fourier transform and its inverse while I_s and I_t denote the target and source images respectively, and M is mask that governs the radius of low-frequency area.

Self-Training with DACS: DACS uses pseudo-labels created by a source-trained model in order to further improve domain adaptation to the target domain images. Supervised learning with such pseudo-labels is done without ground-truth labels. DACS also uses ClassMix, which combines a labeled source image with a pseudo-labeled target image through sampling object-specific region, allowing domain mixing on a class level.

The final training dataset consists of:

- Source images and labels from GTA5
- Pseudo-labeled images from Cityscapes, predicted using the FDA-trained BiSeNet
- Mixed images and masks produced using ClassMix

This dual approach of adaptation (frequency level FDA and pixel level DACS) allowed considerable performance improvements.

4. Results and Discussion

4.1. Classic Training (Baseline on Target Domain)

In order to come up with the upper bound of segmentation performance in the absence of domain adaptation training DeepLabV2 and BiSeNet are trained in fully-supervised manner on the Cityscapes dataset. The step enables a performance benchmark, reflecting anticipated accuracy and efficiency of models with no gap in domains between training and test samples.

DeepLabV2 Upper Bound: The model DeepLabV2 is a high-capacity model with high segmentation accuracy which is trained on Cityscapes training set and tested on validation set. The metrics are provided in table.

Metric	Value
mIoU	57.42
FLOPs	92.12 GFLOPs
Latency (ms)	67.44
Parameters (M)	61.99

Table 1. Performance of DeepLabV2 trained on Cityscapes

BiSeNet Trade-off: BiSeNet is chosen due to its real time inference possibility. Trained on Cityscapes it showed decent segmentation accuracy with low latency and computing requirements.

Metric	Value
mIoU	54.08
FLOPs	82.957 GFLOPs
Latency (ms)	13.38
Parameters (M)	23.231

Table 2. Performance of BiSeNet trained on Cityscapes

These observations verify the trade-off as expected, DeepLabV2 is more accurate as it secures high value of mIoU than BiSeNet but even with the fastest implementation is not applicable to real-time models whereas BiSeNet is a lightweight model that has similar accuracy.

4.2. Domain Shift Analysis

The BiSeNet is trained on a synthetic dataset GTA5 and tested on Cityscapes validation set in order to analyse the domain shift problem in semantic segmentation. The results in this Table highlight a significant performance drop compared to the baseline table due to the domain shift between synthetic and real data.

Class	IoU (%)
road	22.10
sidewalk	6.15
building	61.11
wall	6.77
fence	1.97
pole	9.14
traffic light	4.16
traffic sign	5.99
vegetation	53.09
terrain	3.54
sky	42.36
person	38.78
rider	0.00
car	26.81
truck	2.80
bus	0.97
train	0.00
motorcycle	7.08
bicycle	0.00
mIoU	15.41

Table 3. BiSeNet trained on GTA5 and tested on Cityscapes.

The drastic drop in segmentation accuracy can be attributed to several factors:

- **Visual Disparity:** GTA5 images are synthetic as compared to real-images in Cityscapes as they differ in terms of texture, illumination and shape of objects.
- **Lack of Target Distribution Exposure:** The model was never trained on real-world images which results in poor generalization.
- **Label Style Variance:** Variation in quality of annotation and the annotation mask boundaries convey a performance mismatch.

Data Augmentation to Mitigate Domain Shift: To reduce the domain gap, basic data augmentations are introduced during training. Specifically, horizontal flip (Aug. 1), color jitter (Aug. 2), and their combination are applied with 0.5 probability.

Horizontal Flip Augmentation: The technique is based on a simple expansion of spatial orientation which is intended to increase model generalization. The increment is however minimal as it can be seen in the following table.

Class	IoU (%)
road	8.39
sidewalk	10.40
building	55.77
wall	13.54
fence	8.44
pole	4.36
traffic light	5.91
traffic sign	3.92
vegetation	70.44
terrain	5.92
sky	66.95
person	32.06
rider	0.00
car	22.15
truck	1.44
bus	2.90
train	0.00
motorcycle	0.00
bicycle	0.00
mIoU	16.45

Table 4. BiSeNet performance with Horizontal Flip augmentation (Aug. 1) on GTA5 → Cityscapes.

Color Jitter Augmentation: Introducing color jitter data augmentation technique helped to increase the generalisation of the model compared to the horizontal flip, especially in texture-heavy classes like vegetation and car. Below table presents the per-class IoUs.

Class	IoU (%)
road	76.10
sidewalk	29.20
building	75.23
wall	21.36
fence	16.78
pole	23.08
traffic light	16.12
traffic sign	13.91
vegetation	77.30
terrain	24.35
sky	79.41
person	43.08
rider	4.64
car	48.37
truck	15.80
bus	12.18
train	2.81
motorcycle	6.74
bicycle	0.0000
mIoU	30.87

Table 5. Per-class IoU (Color Jitter Augmentation) — BiSeNet trained on GTA5, evaluated on Cityscapes

Combined Augmentation (Flip + Jitter): The combination yields the best results among the augmentation strategies, significantly boosting classes like *road*, *building*, and *car*.

Class	IoU (%)
road	71.81
sidewalk	19.36
building	74.84
wall	26.07
fence	14.29
pole	24.28
traffic light	24.98
traffic sign	16.16
vegetation	79.58
terrain	23.30
sky	78.03
person	40.45
rider	3.82
car	50.56
truck	5.90
bus	3.24
train	4.86
motorcycle	6.70
bicycle	1.17
mIoU	29.97

Table 6. Per-class IoU (Flip + Jitter Augmentation) — BiSeNet trained on GTA5, evaluated on Cityscapes

While these simple augmentations offer improvements, they are observed to be sufficient to resolve domain shift.

From the results, it is observed that the combined augmentation techniques proved to be a better strategy for classes like train, motorcycle and bicycle. Additionally, the results also reflect the need for more advanced domain adaptation techniques like Fourier Domain Adaptation (FDA) and self-training approaches (DACS), which are explored in the next section.

4.3. Domain Adaptation: Fourier and DACS Strategies

The significant decline in performance, which is witnessed in the last step, clearly indicated the difficulty associated with domain shift. Hence, Domain adaptation methods are used to adapt the two domains by aligning either the image level or the feature level in order to reduce this problem.

In the current project, we used the image-to-image domain adaptation strategy. In particular, we applied **Fourier Domain Adaptation (FDA)**, and **Domain Adaptation via Cross-domain Mixing (DACS)**. These two methods aim to reduce the domain gap between the synthetic GTA5 dataset (source) and the real Cityscapes dataset (target) without needing any of the target domain labels.

Fourier Domain Adaptation (FDA): FDA works on the frequency domain since it maps low frequency information (global structure and appearance) of the target image to the source image whilst preserving the semantic content of the source.

Domain Adaptation via Cross-domain Mixing (DACS):

The DACS model uses the pseudo-label model, which a pre-trained model creates on the target domain. With a class-mixing approach, the source labels are then mixed with these pseudo-labels, in effect forming hybrid training batches containing both real and synthetic appearances.

The DACS pipeline used in this project involved:

- Training BiSeNet on FDA-adapted GTA5 data
- Generating pseudo-labels for target domain (Cityscapes)
- Re-training using mixed batches from both domains

Implementation Details: The augmentation setup which was observed to be the best in the preceding step (Flip + Color Jitter) is employed during FDA, as well as DACS training. Using backbone ResNet-18, the models are trained over 50 epochs. The metrics are reported in Tables as below.

Class	IoU (%)
Road	71.53
Sidewalk	25.06
Building	67.71
Wall	15.36
Fence	12.14
Pole	21.71
Traffic Light	16.41
Traffic Sign	12.93
Vegetation	79.07
Terrain	17.43
Sky	70.83
Person	36.45
Rider	0.29
Car	47.13
Truck	8.06
Bus	0.47
Train	5.19
Motorcycle	4.78
Bicycle	0.02
Mean IoU	26.98

Table 7. Per-class IoU for BiSeNet with Fourier Domain Adaptation (FDA) on GTA5 $\rightarrow$ Cityscapes

Class	IoU (%)
Road	86.92
Sidewalk	16.52
Building	57.81
Wall	7.16
Fence	11.77
Pole	16.09
Traffic Light	8.06
Traffic Sign	6.84
Vegetation	66.26
Terrain	12.73
Sky	50.30
Person	33.31
Rider	3.52
Car	72.31
Truck	11.48
Bus	4.44
Train	0.00
Motorcycle	6.23
Bicycle	0.00
Mean IoU	24.83

Table 8. Per-class IoU for BiSeNet with DACS on GTA5 $\rightarrow$ Cityscapes

FDA vs. DACS Performance: Fourier Domain Adaptation (FDA) seems to be useful at alleviating the visual differences between these two domains (synthetic (GTA5) versus real (Cityscapes)), hence it can be used to adapt at the appearance level and eventually improves the domain adaptation. Here, the FDA manages to implement semantic target-domain supervision and to increase the perfor-

mance. DACS, on the other hand, comparatively does not improve domain adaptation although it consists of mixing source data and pseudo-labeled target-domain data in a spatial mixing manner. The combined result demonstrates the usefulness of pseudo-label based self-training and image-style translation techniques.

4.4. Proposed Extension: Hybrid Loss with CrossEntropy and Dice

Although the DACS approach had already demonstrated notable advantages of minimizing the gap between the baseline and the methodologies of FDA and other approaches, this method still had limitations in effectively cutting the minor or boundary classes like bicycle, fence, and pole. To address this issue, we offered an extension to it with the inclusion of the hybrid loss function that attributes the superiority of CrossEntropy Loss and Dice Loss.

Motivation: CrossEntropy Loss becomes popular in semantic segmentation because it allows to perform multi-class pixel-wise classification. Nevertheless, due to treating pixels separately, it can be bad with highly skewed classes. While, Dice Loss is proposed and is tailored at maximizing overlap of the predicted and true segments more directly. With the combination of two, it is projected to lead to improved generalization especially to struggling classes.

$$\mathcal{L}_{\text{hybrid}} = \mathcal{L}_{\text{CE}} + \lambda \cdot \mathcal{L}_{\text{Dice}} \quad (1)$$

Class	IoU (%)
Road	88.75%
Sidewalk	33.53%
Building	70.65%
Wall	20.21%
Fence	16.08%
Pole	21.74%
Traffic Light	20.19%
Traffic Sign	10.34%
Vegetation	71.40%
Terrain	13.73%
Sky	49.09%
Person	37.50%
Rider	0.00%
Car	71.37%
Truck	10.69%
Bus	1.02%
Train	0.00%
Motorcycle	0.00%
Bicycle	0.00%
Mean IoU	28.23%

Table 9. Per-Class IoU (%) and Mean IoU on Cityscapes Validation Set

The model exhibits significant segmentation performance for prominent and frequently occurring classes such as *building* (70.65% IoU) and *car* (71.37% IoU), suggesting that the DACS training approach, enhanced with Dice loss, effectively captures large-scale structures. Other than that, the model also moderately performs with smaller or less frequent categories, including *pole*, *person*, and *traffic sign*, which all an improved IoU scores.

5. Conclusion

Overall, this report methodologically explored real-time domain adaptation in semantic segmentation of synthetic (GTA5) to real (Cityscapes) urban environments with BiSeNet and illustrated the extent of domain shift as well as the performance of domain adaptation solutions. The experiments validated that, despite the fact that BiSeNet can be trained on the target domain to achieve high real-time performance, the mIoU decreases drastically (approximately 54 percent to 15 percent) when only synthetic data is used, which demonstrates that adaptation is required. Simple data augmentation yielded marginal increases in gains whereas Fourier Domain Adaptation (FDA) reduced appearance differences and enhanced cross-domain performance considerably. The DACS self-training additionally relied on pseudo-labels and cross-domain mixing, and the suggested hybrid CrossEntropyDice loss was able to alleviate the issue of class imbalance, bringing further improvements to smaller and borderline classes. In general, the last pipeline demonstrates that frequency-level adaptation, pseudo-label-based self-training, and loss-level balancing can significantly reduce the synthetic-to-real gap and remain practical in the deployment of semantic segmentation models in autonomous driving contexts in the real world.

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